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PALAEOMAGNETISM, PETROLOGY AND GEOCHRONOLOGY OF TERTIARY MAGMATIC AND SEDIMENTARY UNITS FROM IRAN

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ABSTRACT

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An amount of 160 samples from igneous and sedimentary formations of Palaeocene or Eocene age was collected in Iran, northeast of the Zagros belt. Palaeomagnetic analyses, petrological studies, and K–Ar dating indicate Palaeocene and Eocene palaeomagnetic directions in the Zabol–Baluch region (East Iranian Range) and show that the area remained approximately stable with respect to the Eurasian and African plates from the beginning of the Cainozoic Era. Conversely, large post-Eocene deformations occurred in Central Iran. These large rotations are both clockwise and anti-clockwise. They might have resulted from lateral shears between Arabia and northeast Iran. The Central Elburz underwent a large clockwise rotation which is probably late Oligocene in age. As a whole, our results show that the so-called "Persian Plate" did not exist during the Cainozoic Era.

INTRODUCTION

In the present configuration of the six principal lithospheric plates (Le Pichon, 1968) Iran is found to be in a collision zone between the Eurasian plate to the north, the African plate to the southwest, and the Indian plate to the southeast. The geology of the region shows that Iran is divided among the four domains constituting the Alpine chains of central Asia (Stöcklin, 1977), namely: (1) a southern domain with the Zagros, (2) a central ophiolitic domain represented by the two exterior and

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interior ophiolitic belts (outcropping to the southwest and northeast of the Zagros Main Thrust respectively), (3) a central domain including Central Iran s.l. and the Elburz; and (4) a northern domain represented here by the south Caspian Sea depression and the Kopet Dag.

In regions of such geological complexity, palaeomagnetic analysis must be brought in progressively, beginning with studies of the most recent periods. In effect, reconstitution of previous movements is only possible after having taken into account the effects of the most recent deformations which in the case of Iran are of Alpine age. The late Cretaceous–Eocene period marks a turn in the Alpine history of the region. One assigns to this period the closing of the upper Zagros Alpine ocean (Berberian and King, 1981) dated to the end of the Cretaceous (Takin, 1972) and that of the central Iran oceanic basins the last vestiges of which would be of Eocene age (Takin, 1972). Since that time, Iran has been situated in the continental zone of collision between, essentially, the Eurasian plate to the north and African, and then Arabian plate to the south (McKenzie, 1970). During only the last 10 m.y., this tightening has resulted in a shortening of about 200 km (Le Pichon et al., 1973, p. 96).

The region is therefore particularly interesting in that it provides an opportunity to study compressed emerged zones between lithospheric plates. With the view toward enhancing understanding of this region, we have examined the palaeomagnetics as well as the petrology and geochronology of the reputedly Eocene volcanism. Red sedimentary formations which appeared sufficiently well dated also were sampled, and their magnetism studied.

The age of the magmatism corresponds fairly well to the first phases of the continental compression period. In fact, the omnipresence of this magmatic phase to the northeast of the Zagros allows it to cover the entire hypothetical Persian plate proposed by some authors (Morgan, 1972; Le Pichon et al., 1973, p.83). We may thus hope, through a wide-reaching palaeomagnetic study of the Eocene, to show both a movement of the whole of this region with respect to the major neighbouring plates, and eventual internal deformations of Iran itself.

Previous to our work, a preliminary palaeomagnetic study of Palaeogene rocks had been briefly reported by Soffel et al. (1975), without precise specification of the sampling sites in central Iran. In parallel with our study, Bina (1979) has made a palaeomagnetic analysis of Eocene rocks from the Ghom region, and Conrad et al. (1981) a study of the Palaeocene volcanism of the area around Bam. The results we are reporting in the present paper are relative to four areas: Central Iran s.s., Lut block, East Iranian range, and Elburz chain (Fig. 1).

METHODS OF STUDY IN THE LABORATORY

Petrology and magnetic mineralogy

An amount of 152 thin sections and 24 polished thin sections was studied. The main object of this work was to determine the petrographic nature, eventual degree

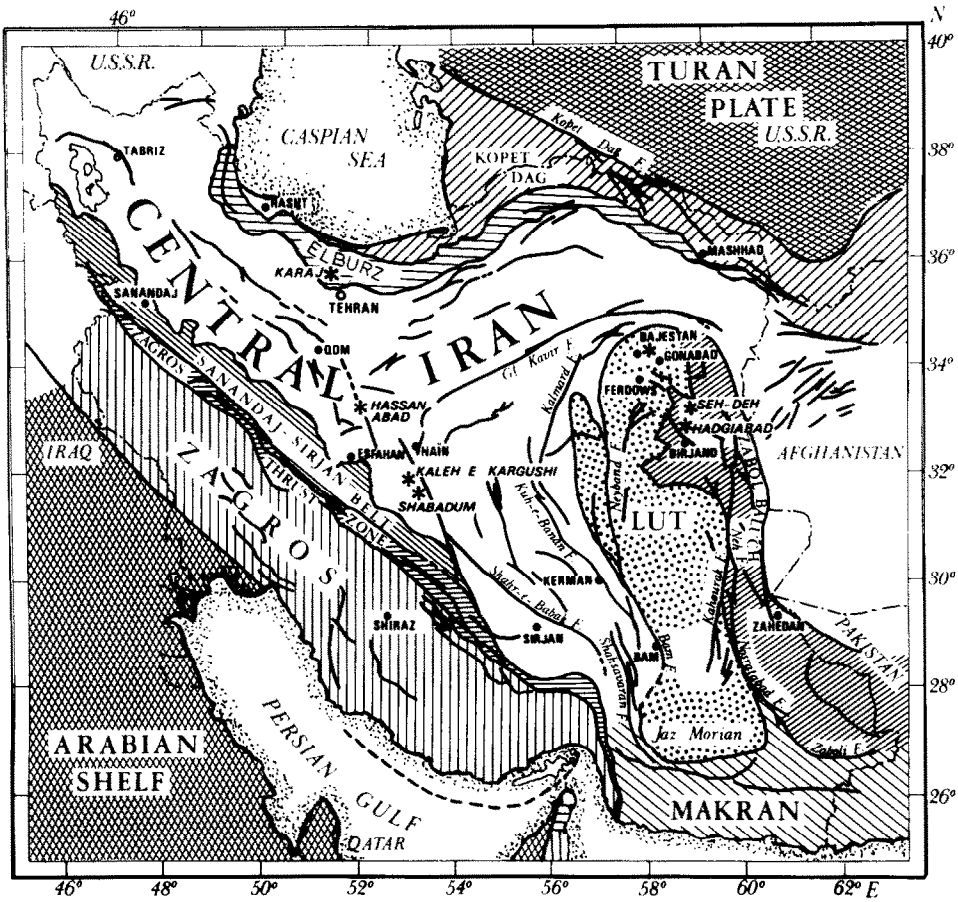


Fig. 1. Main structural areas in Iran and location of sampling sites (asterisks).

of weathering/alteration or metamorphism, and the composition of magnetic phases for each of the eruptive formations examined.

Geochronology

The analytical techniques have been described elsewhere (Cantagrel, 1973; Cantagrel and Baubron, 1983). The material used is represented by "whole rock" samples of 1–2 g and of a particle size varying from 0.3 to 1.0 mm. The argon is extracted by fusion after degassing at moderate temperature (150°C) in high vacuum and dosed by isotopic dilution (spike 38 Ar; $0.8 \cdot 10^{-11}$ mole TPN per analysis) with a modified MS 10 mass spectrometer. The potassium is analysed by atomic absorption spectrometry. The converted ages are calculated with the following constants: 40K = 0.01167 atom.%;

$\lambda(^{40}K\beta) = 4.962 \cdot 10^{-10}$ yr^{-1}. $\lambda(^{40}Ke) + \lambda'(^{40}Ke) = 0.581 \cdot 10^{-10}$ yr^{-1}. The analytical errors were calculated according to Dalrymple's and Lanphere's method (1969).

Magnetism

With the exception of the Seh-Deh redbeds, the magnetic measurements were made at the Laboratoire de Géomagnétisme, Saint-Maur, using the Thellier inductometer (1967) or a JR-4 spinner magnetometer. After determination of their viscosity index (Thellier and Thellier, 1944; Prévot, 1981), the rocks were progressively demagnetised, mainly by alternating magnetic fields (up to 230 mT peak field), and sometimes by heating, in zero field, up to 650°C; other specimens were submitted successively to the two above treatments. The Seh-Deh sediments were studied at the University of Tehran, using an inductometer and AF apparatus built by Schonstedt. In addition, some thermal analyses of these samples were carried out at Saint-Maur.

CENTRAL IRAN s.s.

Geology

Strictly speaking, Central Iran corresponds to the roughly triangular domain delimited by the Lut block to the east, the Elburz chain to the north, and the Sanandaj–Sirjan chain to the southwest (Stöcklin, 1968).

If the Palaeogene of Central Iran s.s. is essentially characterised by a sub-aerial volcanism, during the Neogene there is in addition the establishment of plutonic batholiths sometimes as hypo-volcanic pockets. The whole of this Tertiary magmatism is of calc-alkaline type, with the exception of some potassic episodes marked petrographically by a spectacular shoshonitic series (porphyritic rocks rich in analcitic pseudo-leucite megacrystals). The principal types of calc-alkaline lavas are the andesites, dacites, rhyodacites, and rhyolites.

Amidi (1975) distinguished five different phases of magmatic activity on the basis of palaeontological and stratigraphical evidence:

(1) an Eocene volcanic phase, mostly andesitic to rhyodacitic, beginning with the Lutetian and concluding probably with the beginning of the Oligocene;

(2) a volcanic phase of Oligocene age, essentially rhyolitic. The lavas of this episode transect the Eocene volcanites and are metamorphosed by Miocene plutonic intrusions;

(3) a mainly volcano-sedimentary phase of andesitic magmas, which form a period extending from the Oligocene to the Miocene;

(4) the middle Miocene is characterised by plutonic intrusions, with at first the establishment of the most basic facies (gabbros and diorites), and then the upwelling of more differentiated magmas in sub-volcanic intrusions (granites and grano-diorites of veined and annular structure);

(5) lastly, in the Pliocene, an acid volcanism is represented by pyroclastic sills and extrusions.

The Palaeogene magmatic formations may have been subjected to a retromorphosis, more or less according to the region, interpreted by Amidi (1975) as a static regional metamorphism in green-schist facies, or even amphibolite facies in the Kuh-e-Karkas to the north of Nain. Amidi (1975) has shown that this metamorphism was developed in several phases and that it could have been cogenetic with successive magmatic pulses from the Cretaceous to the Miocene. It would have been provoked by substantial invasions of fluids percolating through the carapace of volcanic material. An important fracturing appears to have controlled the distribution of the effects of this metamorphism. The temperature conditions remained relatively low, certainly below 400°C. The mobilisation of elements such as Ca, Fe, Na, K, and sometimes Si, was widely facilitated. Only the end of this phenomenon is clearly marked by the important phase of plutonic activity radiochronologically dated to the Miocene and whose products are exempt from all transformation. This metamorphism is obviously an element of primary importance for the interpretation of geochronological and palaeomagnetic data. Neogene rocks are exempt from this metamorphism. The metamorphic episode would thus be situated at the Oligocene–Miocene boundary. Most of these formations have been moved to a greater or lesser degree, either because of major folding in neighbouring regions, or as a result of a widely developed tectonic breaking. The series of tectonic phases occurred during the Cainozoic in Central Iran s.s., some major movements being assigned to the late Eocene and late Miocene (Berberian and King, 1981). We have sampled three sites: Kuh-e-Shabadum, Kuh-e-Kaleh-e-Kargushi, and Hassan-Abad.

Kuh-e-Shabadum series

The site at Kuh-e-Shabadum ($\lambda = 32°08'30''N$, $\Phi = 53°15'30''E$) consists of a sequence of andesitic tuffs and flows (Amidi, 1975) with some interspersed calcareous layers of Eocene age (Fig. 2). The monoclinal structure corresponds to $p = 40°$ and $\beta = 210°$ where p is the angle of dip and β the azimuth of the true dip direction measured from 0°to 360° clockwise. Palaeomagnetic samples were taken from six flows of tuffs (Fig. 2).

Petrography

The volcanism of the Kuh-e-Shabadum, as well as that of the Kuh-e-Kaleh-e-Kargushi, has all the characteristics of aerial eruptions: columnar jointed flows interstratified with projections of volcanic bombs and lapilli of vesicular structure. These projection horizons show red-partings produced by thermal metamorphism at the base of the flows.

The Kuh-e-Shabadum lavas all present an important retromorphosis due to the static metamorphism shown by Amidi (1975). These rocks show no trace of

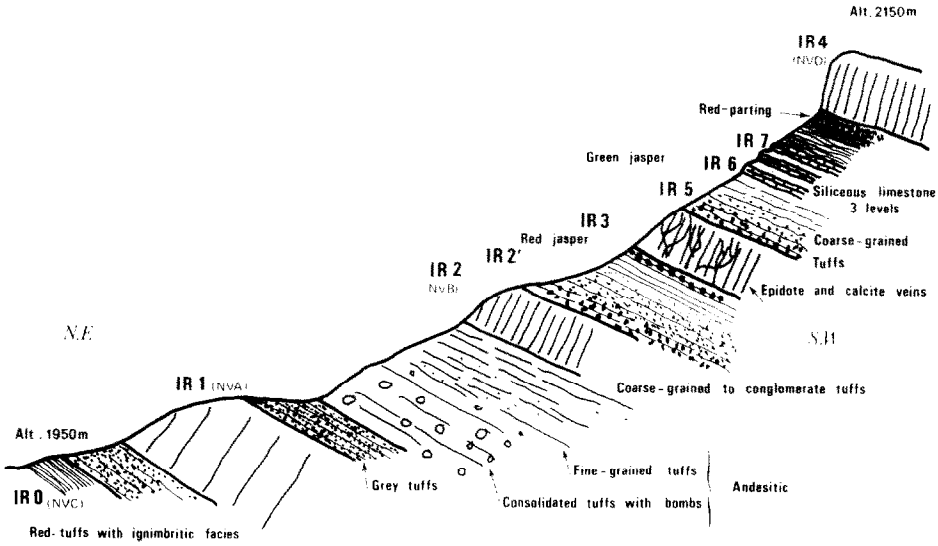


Fig. 2. NE–SW geological section of Kuh-e-Shabadum. Petrological and geochronological sampling is indicated by notation IR. Palaeomagnetic labelling, if any, is given within brackets.

microstructural deformation, the whole of the series having been simply tilted. The secondary paragenesis is of uncoloured phyllites of sericitic type, with chlorites, epidotes, calcite, and haematite. The abundant calcite exists both in a diffused state throughout the rock, and also as secant veinlets or vesicule fill.

Amidi defines these rocks as augite andesites; the analysis which he gives (Amidi, 1975, p. 56, No. 103) allows the calculation of the CIPW parameters: II, (4) 5, '4, 4. The qualitative mineralogical composition is as follows:

Phenocrysts: An_{35-45} plagioclases, clinopyroxenes (augite) and pseudo-morphosis (of olivine?).

Mesostasis: two feldspars (including the An_{25} plagioclase), clinopyroxenes, and opaque oxides.

Secondary minerals: haematite, chlorite (penninite), epidote, calcite.

Geochronology

Two samples (R9062, R9064) form these andesitic flows (IR_2 and IR_4) gave K–Ar ages of around 26 m.y. The observation in these lavas of metamorphic parageneses prohibits the interpretation of this age as that of the volcanism but rather as that of the late Oligocene post metamorphic cooling, in good agreement with geological data (Table 1).

Magnetism

The most frequent magnetic mineral is haematite, resulting either from the martitisation of titanomagnetite, or from the oxidation-replacement of silicates. This

TABLE 1

Geochronological results

site	Petrographic type	K–Ar reference *	Palaeomagnetic reference **	K (%)	40Ar ng/g	Ar atm. %	Age *** (Ma)	Comments
North Karaj	diorite	R 7025 (KJ_1)	EKJV 1–5	2.16	4.08	55.6	27.0 ± 1.3	age of intrusion
South Karaj	diorite	R 7026 (KJ_2)	EKJV 6–10	2.95	7.00	45.5	34.0 ± 1.3	rocks mostly granular possibility of hetero-geneity at sampling
Kuh-e-Shabadum	andesite	R9062 (IR_2)	NVB 1–7	1.73	3.05	40.9	25.3 ± 0.9	age of metamorphism
	andesite	R 9064 (IR_4)	NVD 1–7	4.46	8.28	20.0	26.5 ± 0.7	age of metamorphism
Kuh-e-Kaleh-e-Kargushi	absarokite	R 9061 (IR_{12})	NG 1–7	2.03	3.34	57.6	23.5 ± 1.2	age of metamorphism
		R 9063 (IR_{22})	(1)	0.331	0.478	78.6	20.7 ± 2.0	age of metamorphism
Hassan-Abad	rhyodacite	R 10452 (IR_{64})	NTD 1–6	1.85	2.38	47.7	18.5 ± 0.7	age of volcanism
		R 10453 (IR_{63})	(2)	2.19	2.93	47.4	19.2 ± 0.7	age of volcanism
Bajestan-Gonabad	andesite with orthopyroxenes	R 7030 (TAN_3)	BT 23–26	1.97	7.50	28.5	54 ± 2	
		R7029(TAN_2)	BT 6–12	1.59	6.78	29.3	61 ± 2	secondary alteration
	andesite with orthopyroxenes	R 7028(TAN_1)	BT 1–5	1.74	7.61	47.1	62 ± 2	minimum age
Hadgiabad	andesite	R 7027 (BRD_1)	BJB 6-13	1.93	3.01	65.2	22.5 ± 1.5	secondary alteration; minimum age

* Between parentheses are shown, in the K–Ar reference column, the references of the block taken in the field.

** (1) Rhyodacite top of the series, not sampled for palaeomagnetism, (2) rhyolite top of the series; not sampled for palaeomagnetism.

*** The ages are calculated with the following constants: $^{40}K = 0.01167$ atom.%; $\lambda(^{40}K\beta) = 4.962 \cdot 10^{-10}$ yr^{-1}; $\lambda(^{40}Ke) + \lambda'(^{40}Ke) = 0.581 \cdot 10^{-10}$ yr^{-1}.

is the only magnetic mineral recognised in certain flows (NVC, NVA, NSF). The summital flow (NVD) on the other hand contains only homogeneous titanomagnetites. In the other units, titanomagnetite and haematite are both present. It is important to note the presence of partially maghemitised titanomagnetites in one of the flows (NVB). Since the titanomaghemites decompose at temperatures above 350°–400°C (Readman and O'Reilly, 1970), the metamorphism temperature must not have passed this interval, as is also indicated by petrological data. Moreover, it must be stressed that the haematite recrystallisation does not appear to be linked with the appearance of the metamorphism minerals. The magnetisation carried by the haematite therefore was probably acquired during the deuteric alteration. Magnetic tests are, however, necessary in order to verify that no thermoremanent magnetisation was acquired during the cooling which followed the metamorphism phase.

The magnetic analysis of these units, like all those obtained in Central Iran, was found to be delicate because of the importance of parasite magnetisations. These were due to the fact that many of these samples, which were sawn or cored before magnetic study (in contrast to those from the Elburz– and Zabol–Baluch), are very sensitive to magnetisations provoked by sawing and coring (Lauer, 1978).

After progressive alternating field treatment, the individual directions are satisfactorily grouped in each flow (see Fig. 4a), setting aside flow NVB whose mean direction is poorly defined, and flow NVD, whose primary magnetisation direction could not be defined. Five samples whose magnetisation is carried by haematite were subjected to heating in zero field, after alternating-field treatment. This treatment could not be applied to rocks containing titanomagnetite, which all carried important spurious magnetisations due to cutting or coring, and which acquired besides strong anhysteretic remanent magnetisations during alternating-field treatment. The magnetisation of haematite rocks is very resistent thermally (40–100% of the moment subsists at 470°C) and its direction remains constant during treatment. It appears that the heating at around 350°C which would have accompanied the metamorphism, has not effectively modified the direction of the magnetisation. This demonstrates both the high level of haematite blocking temperatures, and doubtless also the fact that the metamorphism is antetectonic.

The mean direction of magnetisation which may be associated with the five Eocene formations of Kuh-e-Shabadum studied here, is (Table 2): $I = -50.4°$, $D = 226.9°$, with $k = 15$ and $\alpha_{95} = 16°$.

The Kuh-e-Kaleh-e-Kargushi series

The Kuh-e-Kaleh-e-Kargushi series, geographical neighbour of the preceding series ($\lambda = 32°12'00''N$, $\Phi = 53°11'15''E$), has a similar structure and dip (Fig. 3). It consists, however, of lavas very different in nature and facies from those of Kuh-e-Shabadum. According to Amidi (1975), this is a shoshonitic association from

TABLE 2

Mean magnetisation directions of the units studied, after magnetic cleaning and dip corrections *

Structural domain	Site	Unit	N	v (%)	I (°)	D (°)	k	α_{95} (°)
Central Elburz	Karaj (sill)	EKJV 1–10	10	4.0	56.0	65.0	(1)	(1)
Central Elburz	Karaj (tuff)	EKJT 1–5	5	5.5	59.6	48.2	206	4.4
Central Iran s.s.	Kuh-e-Shabadum	NVC 1–6	6		−43.7	245.2	202	4.0
		NVA 1–6	7		−44.3	251.7	74	6.2
		NVB 1–7	4	3.1	−28.4	205.6	34	12.0
		NSF 1–6	7		−59.7	214.4	231	3.5
		NVE 1–7	6	1.8	−67.3	211.8	168	4.4
Central Iran s.s.	Kuh-e-Kaleh-e--Kargushi	NG 1–7	5	2.1	−32.9	236.6	49	9.0
		NH 1–6	7	2.2	−27.8	247.8	102	5.2
		NI 1–6	8		−33.2	236.6	61	6.4
		NJ 1–6	6		−36.3	250.0	37	9.6
		NK 1–7	10		−40.0	233.2	122	4.0
Central Iran s.s.	Hassan-Abad	NTD 1–6	5	2.3	49.0	344.5	28	12.0
Lut Block	Bajestan–Gonabad	BT 23–26	4	3.5	−58.6	160.1	218	4.7
		BT 19–22	3	4.2	−49.3	175.7	110	7.7
		BT 13–18	6	22.3	−50.8	157.0	46	8.4
		BT 6–12	7	2.7	−56.2	186.4	80	5.9
		BT 1–5	4	3.5	−53.8	152.0	326	3.9
Zabol–Baluch	Seh-Deh	BJC 1–9	9		59.6	343.3	59	6.0
Zabol–Baluch	Hadgiabad	BJB 1–15	7		−30.0	151.1	77	6.0

* N —number of samples and/or specimens used for the calculation of the mean direction; v —geometrical mean (Prévot, 1981) of the magnetic viscosity coefficient; I and D —inclination and declination of the mean magnetic vector respectively; k —precision coefficient of Fischer (1953); α_{95} —demi-angle at the summit of the 95% confidence cone.

(1) Mean direction defined by the intersection of the demagnetisation circles (precision parameters not calculated).

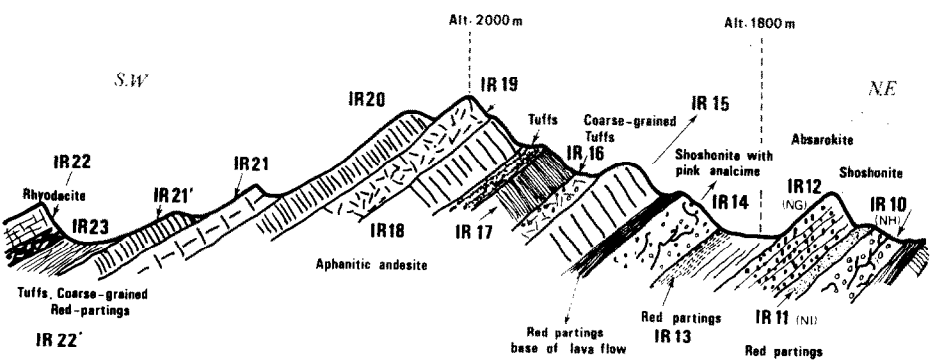


Fig. 3. SW–NE geological section of Kuh-e-Kaleh-e-Kargushi. Sampling labelling as in Fig. 2.

the upper Eocene. Five volcanic or volcano-sedimentary formations were sampled for palaeomagnetism.

Petrography

This shoshonitic series is well characterised by porphyritic lavas rich in phenocrysts: big, greenish-white trapezohedrals, attaining diameters of up to 3 cm. These crystals are abundant and may form up to 40% in volume of the lava. Today, they are either formed of analcite, or of a polymineral association of which analcite is the principal phase; the other associated minerals are: potassic feldspar, albite, prehnite, pumpellyite, epidote, chlorite, thomsonite, and other zeolites. Moreover, a secondary paragenesis of the same composition is frequently observed in all the lavas. At the base of the series, basic terms (absarokites) are characterised by doleritic structures and the presence of a frequently retromorphosed olivine.

The analcite in pseudo-morphoses of old phenocrysts is certainly not a primary mineral in these lavas, both for petrological reasons: incompatibility of the analcite in sub-aerial volcanic conditions (Roux and Hamilton, 1976); and for geochemical reasons: analcite is not a mineral rich in potassium. The microscope and diffractometer studies have shown us that the analcite here is in the form of trapezoidal pseudomorphs alone, or in association with a whole secondary paragenesis already mentioned; but it is also in a quite late position, as vesicule fill and secant veinlets.

Comin-Chiaramonti et al. (1979) have shown that the "automorphic phenocrysts of analcite" from Azerbaijan Eocene lavas have a secondary and metasomatic origin. They result from post-magmatic K–Na ion exchange processes on a leucitic primary phase. We have never found a leucite phase subsisting in pseudo-morphic associations. This ion-exchange, independent of temperature, favoured by low pressures, is even possible under simple conditions of superficial alteration given that the leaching be done by a saline or K/Na solution of not more than 0.3.

An important secondary leaching of the potassium in the samples must therefore be expected, a leaching which could be as easily due to the M-type metamorphism described by Amidi (1975), as to a superficial alteration, or even more probably a result of the accumulated effect of the two phenomena.

Geochronology

A basic sample (absarokite) from the shoshonitic series is dated to 23.5 m.y., and a dacitic sample from the summit to 20.7 m.y. As before, this age marks the end of the metamorphic and/or metasomatic episode (small K content of the absarokite) at the Oligocene–Miocene limit (Table 1).

Magnetism

With the exception of the sedimentary or volcano-sedimentary redbeds (NI and NK) whose magnetisation is carried by haematite alone, the volcanic rocks all contain titanomagnetites. These are sometimes partially maghemitised (NJ) but, for

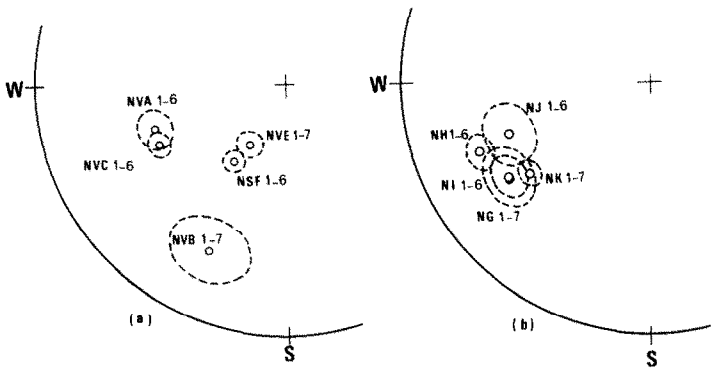


Fig. 4. Naïn (Eocene). Mean direction of magnetization after A.F. cleaning and tectonic correction for individual lava flows or sedimentary units. Circles indicate the α_{95} confidence interval: (a) Kuh-e-Shabadum, (b) Kuh-e-Kaleh-e-Kargushi.

the most part, they are partially martitised (NJ and above all NH), or have been completely oxidised at high temperature (NG: residual magnetite, ilmenite flakes with rutile exsolutions, associated haematite and rutile). All these rocks also include crystallised haematite in partial replacement of silicates. The conclusions are identical to those proposed for the Kuh-e-Shabadum series in terms of temperature of metamorphism (less than 400°C) and the absence of any notable recrystallisation of magnetic minerals during this process.

After alternating-field treatment, the magnetic directions are sufficiently well grouped, with a 95% confidence cone of less than 10°. They are moreover little different from one unit to another, whether volcanic or sedimentary (Fig. 4b, Table 2). The heatings in zero field to which four samples were subjected do not modify the direction found, the magnetisation losing less than 50% of its intensity at 400°C. As for the Kuh-e-Shabadum series, the metamorphism does not appear to have produced a modification of the remanent magnetisation direction, which would thus be characteristic of the upper Eocene (Amidi, 1975).

The mean direction of the upper Eocene field at Kuh-e-Kaleh-e-Kargushi, defined by the mean of the directions of the five formations studied (Table 2), corresponds to: $I = -34.2°$, $D = 241°$, with $k = 112$ and $\alpha_{95} = 5.9°$

Hassan-Abad

The Hassan-Abad series, named after a nearby farm, outcrops 15 km south of Natanz, west of the Ardestan crossroads ($\lambda = 33°24'30''$N, $\Phi = 51°58'00''$E). It is formed of differentiated rhyolitic to rhyodacitic lava flows with pyroclastic tuffs. The structure is monoclinal ($p = 25°$, $\beta = 80°$). The geological map indicates this series to be Eocene. Palaeomagnetic samples were taken from two flows.

Petrography

The lavas are well characterised by their aphanitic texture. The superficial alteration shows up the finely banded facies which is found under the microscope in the form of a clear flow structure. The mesostasis is cryptocrystalline with hyaline, and is made opaque by a great abundance of haematite granules. This richness in haematite is responsible for the general chocolate-brown colouration of these rocks. The phenocrystals, which are rare, and lie flat with the flow, are An_{10-15} plagioclases.

These dacitic lavas do not show the metamorphic paragenesis observed elsewhere in the Eocene series.

This series is made up of pyroclastic flows of ignimbritic type flame structure. We have here a strongly explosive rhyodacitic volcanism probably of post-Oligocene age judging from the absence of the metamorphic episode of the Oligocene–Miocene limit.

Geochronology

The K–Ar measurements of two rhyodacite samples, one from near the IR_{64} base, the other at the IR_{63} summit (Table 1) confirm this conclusion. This thick volcanic series was without doubt rapidly poured out 19 m.y. ago during the lower Miocene.

Magnetism

The magnetic mineralogy is little different from that of the Eocene lavas. Haematite is the most abundant mineral. There are either very fine grains dispersed in the rock (NTE), or recrystallisation minerals (NTC and NTD) formed from silicates or martitised titanomagnetites. In flow NTD, some slightly maghemitised titanomagnetites are also present. The similitude of the magnetic parageneses in this inaltered series, and in the clearly metamorphosed preceding ones, confirms the conclusion proposed above that the metamorphism has not notably modified the nature of the magnetic phases of south Nain flows (cf. the two foregoing sections on the Kuh-e-Shabadum and Kuh-e-Kaleh-e-Kargushi series).

Here again, the parasitic magnetisations, probably due to cutting and coring, are important. The alternating-field treatment did not allow a primary magnetisation direction to be defined for NTE. The results obtained for NTD (Table 2 and Fig. 5) bring a direction for the south-Natanz lower Miocene of approximately $I = 49°$ and $D = 345°$.

THE LUT BLOCK

Geology

In the Cainozoic, the Lut block is a rigid mass surrounded by the more mobile zones corresponding to central Iran and east Iran (Stöcklin, 1968). Following

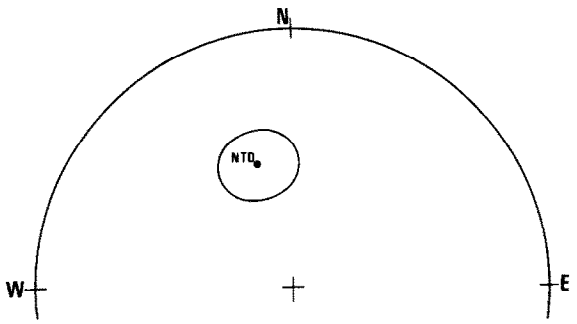


Fig. 5. Hassan-Abad, south of Natanz (Miocene); see caption of Fig. 4.

Conrad and Girod (1977), we will consider three main events in the Tertiary for this block:

(1) between 50 and 30 m.y., emissions of alkaline basalts followed by ignimbritic layers;

(2) around 25 m.y., granitic intrusions of the Kuh-e-Shahsavaran, and new alkaline basalt emissions;

(3) around 20 m.y., a phase of general folding along all the sides of the Lut, while on the block itself the Palaeogene volcanic substratum is only slightly affected tectonically.

The site studied, known as *Bajestan–Gonabad*, is situated on the road linking the two towns ($\lambda = 34°30'N$, $\Phi = 58°17'E$) to the extreme north of the Lut block (Fig. 1). It consists of a vertical sequence of five units (flows and tuffs) reputedly of Eocene age. The monoclinal structure has northeasting dip ($p = 43°$ and $\beta = 53°$).

Bajestan–Gonabad series

Petrography

Three samples were studied, from flows BT1–5, BT6–12, and BT19–22 of the Bajestan–Gonabad volcanic series, at the extreme north of the Lut block. The structure is microlitic, the phenocrysts being plagioclases and orthopyroxenes. The mesostasis is mainly of feldspars, and pyroxene and opaque oxide granulations. These are orthopyroxene andesites. These lavas are very altered, and rich in secondary products: chlorite, sericite, calcite, and haematite. No metamorphism was detected.

Geochronology

Three orthopyroxene andesite samples were used for K–Ar dating (Table 1). The ages measured correspond to the Palaeocene. The presence of hydrothermal alteration secondary minerals in this series makes these ages the minima for the

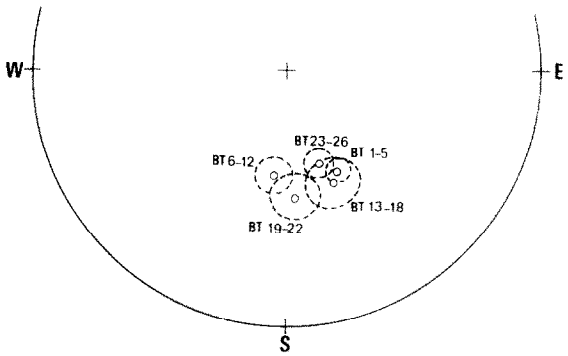


Fig. 6. Bajestan-Gonabad section (Paleocene), northern margin of the Lut block: see caption of Fig. 4.

volcanism. Nonetheless, for reasons of stratigraphical order, the volcanism should not be much earlier.

Magnetism

The primary magnetic minerals have been considerably modified, either during the cooling, or during perhaps more recent hydrothermal episodes. The highest flow (BT1–5) contains mainly primary ilmenite oxidised at high temperature and transformed into ferri-ilmenite with bands of "rutiles" and "pseudobrookite" similar to those described by Haggerty (1976). The haematite, often associated with rutile, mostly results from the partial oxidation of silicates, and the martitisation of titanomagnetites (BT6–12 and BT13–18). Setting aside the summital flow, the original oxides (probably ilmenite and martitised titanomagnetites) of the other flows are entirely transformed (rutile associated with zones of covellite and, more rare, chalcocite) and shot through with vesicules of secondary haematite. These important transformations indicate the hydrothermal conditions during which the characteristic magnetisation was probably acquired.

The magnetic viscosity is weak (Table 2) and the magnetisation directions are well grouped before any treatment. This is probably because the samples had been neither cut nor cored before magnetic analysis. After treatment, the directions, reversed, vary little from one flow to another (Fig. 6), whether they be altered (BT6–12, BT13–18, BT19–23) or not (BT1–5). This would suggest that the hydrothermal episode (dated around 60 m.y.) developed shortly after the eruption. The mean direction of the Palaeocene field in this region, calculated from the means for the five formations studied, is defined by: $I = -54.4°$, $D = 166.2°$ with $k = 78$ and $\alpha_{95} = 7.1°$.

THE EAST IRANIAN CHAIN (ZABOL–BALUCH)

Geology

The East Iranian (Stöcklin, 1968), or Zabol–Baluch chain (Berberian and King, 1981) forms the eastern edge of the Lut block. According to the latter authors, it

would correspond to a north–south arm of the Cretaceous Indian Ocean, closed during the upper Maastrichtian. This period of folding and uplift was followed by the deposition of flyschs, mostly Palaeogene, and by surface or submarine volcanic activity particularly important in the Eocene. The major folding phase of these formations may be situated at the end of the Oligocene (Berberian and King, 1981). Samples were taken from two sites.

Seh-Deh redbeds

The site of Seh-Deh ($\lambda = 33°23'N$, $\Phi = 59°10'E$) is a Palaeocene sedimentary formation built of alternating red sandstone and conglomerate beds. The dip is moderate: $p = 19°$, $\beta = 235°$.

The magnetic viscosity is low and the ARN directions well grouped. The magnetisation is very resistant to alternating fields, but diminishes by heating to 500°C, without change in direction. After treatment (Fig. 7) the direction of the Palaeocene field is given by (Table 2): $I = 59.6°$, $D = 343.3°$, with $k = 59$ and $\alpha_{95} = 6.0°$.

Hadgiabad

The site of Hadgiabad ($\lambda = 33°01'N$, $\Phi = 59°18'E$), situated 20 km north of Birjand, consists of an andesitic volcanic unit overlying well-layered volcano-sedimentary tuffs; the whole may be considered Eocene. The dip is $p = 32°$, $\beta = 105°$. We may note on the 1:1,000,000 scale geological map (sheet 3) a nearby outcrop of a Neogene porphyrite–diorite intrusion, which may have caused reheating of the rocks under study.

Petrography

The sample studied for this flow shows a structure with two crystallisation phases. The phenocrysts are plagioclases, amphiboles (green hornblende) and some biotite

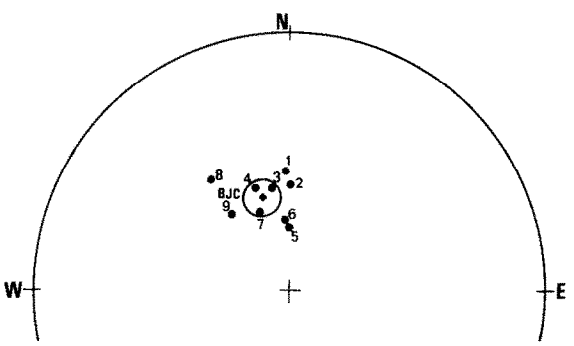


Fig. 7. Seh-Deh redbeds (Eocene), Zabol-Baluch individual directions after cleaning and tectonic correction are indicated by filled circles. The average direction corresponds to the square with the associated 95% confidence circle.

microphenocrysts. The mesostasis minerals are unidentifiable because of the high alteration of the rock. The composition is andesitic. No sign of metamorphism was observed.

Geochronology

The K–Ar age of the flow is around 22.5 m.y. (Table 1). Taking into account the secondary alteration, this age is less than that of the volcanism.

Magnetism

The opaque mineral optically identifiable in the two thin sections of the flow (BJB6–13) is ilmenite, in large crystals, attaining sometimes several hundreds of micrometres. This primary ilmenite presents a partial alteration due to an oxidisation at high temperature, characterised by the appearance of zones of titano-haematite intermixed with ferri-rutile. The thermomagnetic high-field analysis furnishes a roughly reversible curve corresponding to a Curie point around 540°C (Fig. 8). This is a frequent value for slightly substituted magnetites resulting from high temperature oxidation of titanomagnetites. However, this magnetite could not be identified optically, probably because of the smallness of the crystals.

The magnetic viscosity is weak (Table 2). Before treatment, the mean magnetisation direction, after correction for dip, is anormal: $I = -9.3°$, $D = 129°$, with $k = 18$ and $\alpha_{95} = 11.2°$.

For most of the samples, the alternating magnetic field treatment, and still more the thermal treatment, produce a continuous modification of the magnetisation

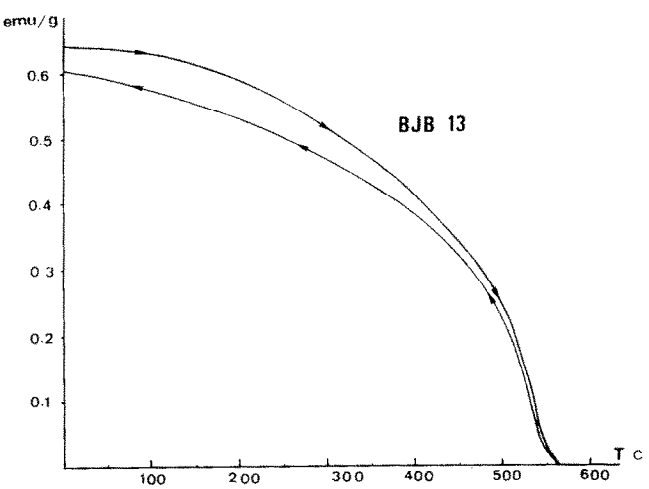


Fig. 8. Sample BJB 13. Thermal variation of strong-field magnetization ($H = 250$ mT) during a single heating-cooling cycle, in vacuum.

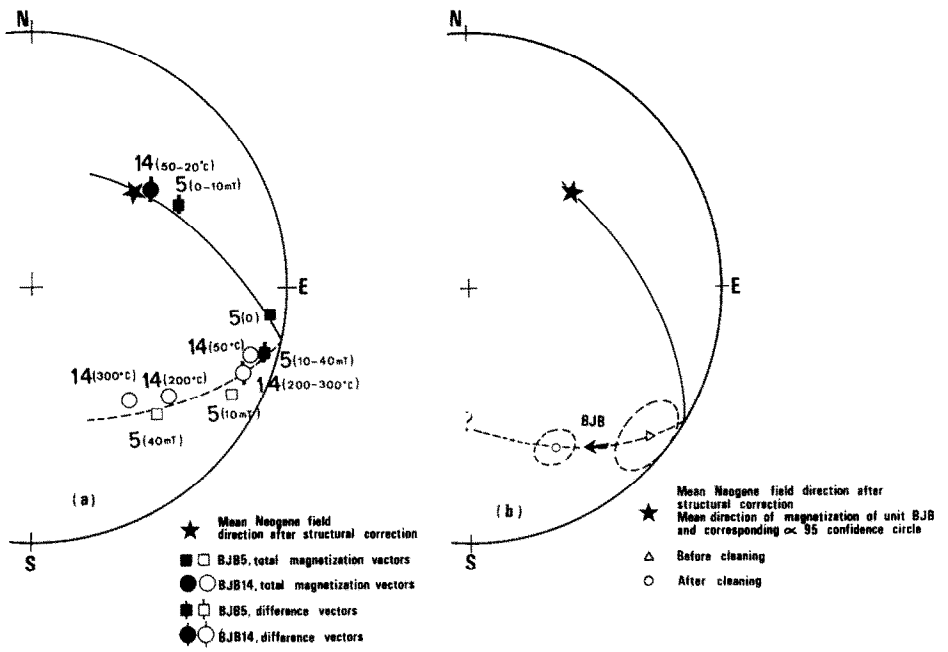


Fig. 9. Hadgiabad (Eocene): a. Thermal demagnetization (sample BJB 14, circles) and A.F. demagnetization (sample BJB 5, squares) of two samples from flow BJB. Some difference vectors are also indicated from tilting. The distribution of directions along a great circle is obvious. At the beginning of the demagnetization processes the difference vectors are quite close to the direction of the Neogene geomagnetic field in this area, suggesting that the flow was partially remagnetized at this time. b. Mean direction of magnetization of the flow, after tectonic correction, before (triangle) and after (circle) A.F. and thermal cleanings. The star corresponds to the Neogene geomagnetic field direction (after unfolding).

direction, particularly clear for samples BJB5 and BJB14 (Fig. 9a). The representative points associated with the remaining magnetisation as well as those corresponding to magnetisations lost during the successive intervals of field intensity or temperature, are situated approximately along a great circle; there is therefore a superposition of two magnetisations.

At the start of the treatment (up to 200°C or 10 mT in the two examples shown), a magnetisation of normal polarity is destroyed. The furthest normal direction on this circle, corresponding to a magnetisation destroyed up to 200°C (sample 14), is defined by $I = 52°$ and $D = 12°$ with respect to the present geographical coordinates. The direction of the Neogene magnetic field in the Hadgiabad region, calculated from the Eurasian and Arabian palaeomagnetic poles at the upper Tertiary (McElhinny, 1973) corresponds to $I = 52°$ and $D = 11°$ with $\alpha_{95} = 5°$. The secondary magnetisation has therefore been acquired during the Neogene, after the folding at the end of the Oligocene, probably during a reheating which accompanied the establishment of the nearby reputedly Neogene batholith. The

K Ar age found for this flow (22.5 m.y.), corresponds in fact to the date of this heating, and of the secondary component.

Seven samples were heated in zero field at higher temperatures: 450°C and 600°C. After treatment, the magnetisation directions, reversed, are fairly well grouped around the following mean value: $I = -30°$, $D = 151.1°$, with $k = 77$ and $\alpha_{95} = 6.0°$.

This direction is without doubt near to that of the primary magnetisation of these Eocene formations. However, the directions obtained at the end of treatment are not necessarily exempt from a secondary component. It is therefore possible that the magnetic declination and inclination are slightly underestimated (Fig. 9b).

THE CENTRAL ELBURZ CHAIN

Geology

The Elburz chain belongs to the central domain of the Asian Alpine chains (Stöcklin, 1977), and may be considered an anticlinorium marginal to Central Iran s.l. which would have begun to appear at the end of the Cretaceous (Stöcklin, 1968).

The Eocene submarine volcanism attains thicknesses of several thousand metres. It is formed mainly of green tuffs interspersed with andesites or basalts as flows, dykes, or sills. They are of middle Eocene age, through to, perhaps, the upper Eocene (Stöcklin, 1972, p.131). They are covered discordantly by a red conglomerate presumed to be of Neogene age (Stöcklin, 1972, p.130) which puts the main folding between the upper Eocene and the end of the Oligocene.

The formation sampled is that of Karaj, the type locality of north Iranian Eocene magmatism. The two sampling sites are on the north and south flanks of a syncline whose frame is formed by a dioritic sill, about 130 m thick, injected into the green tuffs. The northern flank site ($\lambda = 35°57'30''$N, $\Phi = 51°04'55''$E) has a dip of $p = 53°$ and $\beta = 217°$, and the other ($\lambda = 35°54'10''$N, $\Phi = 50°59'00''$E) a dip of $p = 67°$ and $\beta = 9°$. The samples were taken from the sill and, for the north flank, the underlying tuffs, at points less than 10 m from the contact. The intrusion of the dioritic magma has produced contact metamorphism, characterised by lenticular amphiboles in the green tuffs. Given its dimensions, the samples taken had certainly been reheated by the intrusion. The northern outcrop of the sill was sampled at the base of the intrusion, in a more melanocratic facies than that of the southern outcrop, which corresponds to the top of the sill.

KARAJ SITES

Green tuffs

The volcano-sedimentary samples collected just under the diorite present a moderate magnetic viscosity (Table 2) and well-grouped natural remanent magneti-

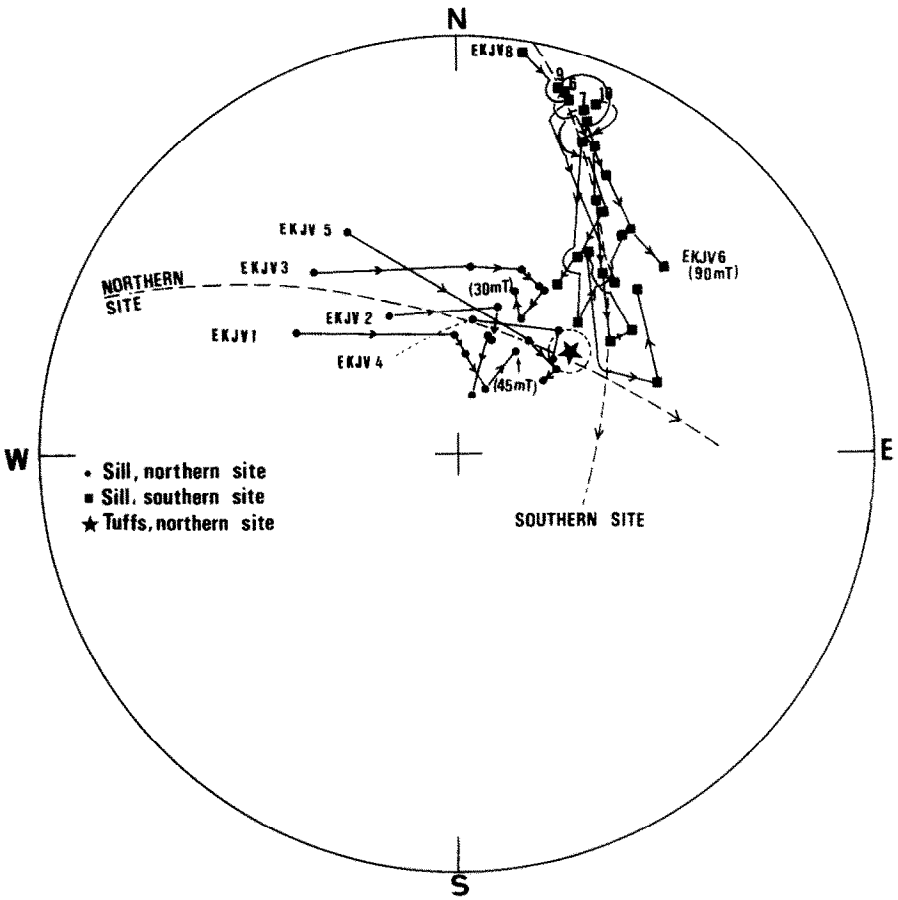


Fig. 10. Karaj (Oligocene). Equal-area projections of remanence directions. Individual samples from the sill correspond to circles (north site) or squares (south site). They have been progressively demagnetized using alternating fields. The star (and associated α_{95} percent confidence interval) corresponds to the average direction of the cleaned remanence of underlying tuffs (north site).

sation (NRM) directions. The magnetisation is resistant to alternating fields, with a median destructive field (MDF) of the order of 50 mT. The mean direction after treatment, very near that of the NRM, is well defined (Fig. 10, Table 2).

Sill

Petrography

Near the roof (leucocratic facies) and walls (more melanocratic facies) of the intrusion, the rock is granular, sub-equigranular with automorphic feldspars. The mineralogical composition is that of a diorite: non-zoned andesine An_{40-50} automorphic plagioclases, clinopyroxenes, a little amphibole (green hornblende) and

biotite, presence of alkali feldspars in a position of terminal crystallisation and association with a little quartz, acicular apatite and automorphic opaque oxides. The clinopyroxene presents the mark of a clear destablisation; it is always surrounded by a ring of chlorite and large xenomorphic crystals of opaque oxides. This chloritisation does not alter the little crystals of biotite; it is probably the mark of late- or post-magmatic deuteric action.

Geochronology

Two samples of this differentiated dioritic sill were used for whole-rock K–Ar analysis. The leucocratic facies (34 m.y.) and the more melanocratic facies (27 m.y.) gave significantly different ages (Table 1) which may be explained either by a heterogeneity in sampling, always possible for largely granular rocks (2–3 mm), or by a slight loss of argon for the more recent date, associated with a certain late magmatic chloritisation. These data would seem nonetheless sufficient to fix an Oligocene age for this intrusion which cuts and metamorphoses the Eocene volcano-sedimentary tuffs.

Magnetism

The opaque minerals, several hundred micronmetres in size, are formed of titanomagnetite and ilmenite (Haggerty, 1976), unaltered and perfectly homogeneous in the samples from the northern site (floor of the sill), and partially oxidised in those from the south (top of the sill). In this case, the titanomagnetite is transformed into pseudobrookite and haematite, and the ilmenite into associations of "rutiles" (rutile and ferri-rutiles) and "pseudobrookite" (pseudobrookite to ferro-pseudobrookite). These oxidations occur at high temperatures, probably above 800°C (Haggerty, 1976). The primary magnetisation of the sill is therefore probably of a thermoremanent nature.

The samples from the northern site have a weak magnetic viscosity, while their magnetisation is not very resistant to alternating fields (MDF between 10 and 15 mT). During treatment, the magnetisation directions change, approximately along a great circle (Fig. 10), tending towards values obtained from the tuffs. The samples from the southern site have an equally low viscosity, but their magnetisation is more resistant to alternating fields (MDF around 30 mT). Here again, the magnetisation directions evolve during treatment, allowing the definition of a second circle of demagnetisation (Fig. 10). The intersection of these two circles, each determined by Halls' method (1976), gives the characteristic magnetisation direction of the sill. It is little different from the mean value found for the tuffs after magnetic cleaning (Table 2). We conclude that the Karaj sill, although intrusive into the Eocene tuffs, is significantly younger and must be attributed to the Oligocene. The magnetisation directions obtained after treatment and correction for dip are in proximity to each other. This magnetisation is therefore clearly anterior to the folding phase which has affected the sill.

We will take the mean of the directions found for the tuff and the sill, $I = 58°$ and $D = 57°$ as the characteristic direction for the Oligocene rocks of Karaj. It is impossible to calculate exactly the error associated with this direction. We may, however, note that the error is lower than that obtained without extrapolation, from the magnetisation directions after magnetic treatment. This maximal value of uncertainty corresponds to $\alpha_{95} = 10°$ ($k = 13$).

DISCUSSION

Two geodynamic models have been proposed to account for the shortening during the Tertiary of the zone situated between the Eurasian and Arabo-African plates. The first supposes that the shortening is absorbed by subduction, from a zone near the Zagros chain, of the Arabian plate under Central Iran s.l. (Nowroozi, 1971). This last region is itself considered a lithospheric plate and so, by hypothesis, undeformable (Persian plate of Morgan, 1972 and Le Pichon et al., 1973, p.83). It is to this eventual Tertiary subduction, which is perhaps still continuing today (Nowroozi, 1971), that is generally attributed the intense Eocene volcanic activity throughout Central Iran (e.g., Vialon et al., 1972; Jung et al., 1976).

However, this interpretation encounters a number of difficulties, in particular because of the age and petrological character of the lavas produced (Berberian and King, 1981). These authors suggest that the shortening is expressed essentially by a thickening of the crust. Severely deformed since the beginning of the collision between the Arabian and Eurasian continental crusts in the upper Cretaceous, the Iranian crust would have approximately doubled in thickness.

Table 3 recapitulates the palaeomagnetic data published up to now for Palaeogene or lower Neogene rocks from Iran. We have added to our results those of Conrad et al. (1981, 1982), which deal with Palaeocene flows of the southern Lut. Despite the very reduced number of oriented samples taken by these authors at each site (Conrad et al., 1982), the mean direction which they obtain by the combination of data from eight different sites would seem fairly precisely defined. For comparison, we also give in this table the mean direction and pole proposed by Soffel and Forster (1980) for the lower Tertiary (Palaeocene and Eocene) of the whole of Central Iran. Finally, we have calculated, from the synthesis of Irving and Irving (1982), the mean expected direction in Iran for the case where the regions studied had remained stable with respect to Eurasia since the lower Tertiary.

Table 3 shows firstly that the reputedly Eocene magmatism, according to the N.I.O.C. 1:1,000,000 geological map of Iran, is distributed in time from the Palaeocene (Bajestan–Gonabad) to the Miocene (Hassan–Abad). Moreover, this spreading out in time is often accompanied by large variations in the mean direction from one site to another. The palaeomagnetic results thus show unequivocally that the tectonic evolution of Iran during the Tertiary forbids us to consider it as a lithospherical plate; the post-collision compression has resulted in considerable

TABLE 3

Mean magnetization directions of the Palaeogene and Neogene sites of Iran and corresponding palaeomagnetic poles *

Structural domains	Site	Age (m.y.)	$\bar{I}$ (°)	$\bar{D}$ (°)	k	α_{95} (°)	$\bar{\theta}$ (°)	$\bar{\Phi}$ (°)	K	A_{95} (°)	References
Central Elburz	Karaj	30.5 ± 3.5	58.0	57.0	–	–	45.0 N	119.0 E	–	–	this study
	Kuh-e--Shabadum	50–55	50.4	46.8	15	16.0	51.1 N	129.2 E	13	17	this study
Central Iran stricto sensu	Kuh-e-Kaleh--e-Kargushi	40–45	34.2	61.0	112	5.9	34.2 N	141.2 E	110	6	this study
	Hassan-Abad	18.9 ± 0.7	49.0	344.5	–	–	76.4 N	48.9 W	–	–	this study
	Qom	40–55	71.0	246.0	25	7.0	16.5 N	18.6 E	..	..	Bina (1979)
	Central Iran	40–65	39.1	330.8	14	7.0	62.0 N	49.4 W	14	7	Soffel and Forster (1980)
Lut Block	Bajestan–Gonabad	59.0 ± 4.0	54.4	346.2	78	7.1	78.6 N	25.5 W	43	9	this study
	Bam	22.5–65	43.0	270.0	11	18.0	12.0 N	9.0 W	–	–	Conrad et al. (1981)
Zabol-Baluch	Seh-Deh	55–65	59.6	343.3	60	6.0	75.0 N	1.8 E	..	..	this study
	Hadgiabad (1)	40–55	30.0	331.0	77	6.0	58.9 N	56.8 W	–	–	this study
From the Eurasian Palaeomagnetic pole		35–65	47.0	17.0	55	5.0	75.0 N	155.0 E	44	6	Irving and Irving (1982)

* The age shown is either the mean K–Ar age which we have determined (in millions of years), with the expected error, or else the time interval attributed by the Geological Time Table (Van Eysinga, 1975) to the geological period corresponding to the stratigraphic age of the volcanism or sedimentation; $\bar{I}$, $\bar{D}$ are respectively: the mean magnetic inclination and declination; k and α_{95}, precision factor of Fisher (1953) and the demi-angle at the summit of the 95% confidence cone for the mean direction of each site; $\bar{\theta}$ and $\bar{\Phi}$: latitude and longitude of the mean virtual geomagnetic pole (VGP) of each site; K and A_{95}, precision factor of Fisher and the error at 95% on the mean of the VGP for each site.

Magnetization directions and polar coordinates correspond here to a normal polarity. The direction of magnetization and the coordinates of the pole correspond to the mean of the directions or poles associated with each volcanic or volcano-sedimentary formation of the site considered (with the exception of the site at Hassan-Abad, where a single flow has furnished a significant mean). For the redbeds, the direction and mean coordinates of the pole are obtained by combination of the results from individual samples. The error on the directions expected for Central Iran (for the latitude 32°N and longitude 54°E) taken from the mean Eurasian pole and the mean pole for Central Iran (Soffel and Forster, 1980) have been calculated from the errors on these poles (Cox, 1970).

(1) Direction approximate because of a strong secondary magnetization not entirely eliminated by magnetic cleaning.

deformation and relative rotations. As expected, the arithmetic mean inclination, calculated from our data (48° ± 6°), is not significantly different from the inclination expected from Eurasian palaeomagnetic data.

The rotations brought to light are of variable direction, and generally of different magnitude with respect to the Eurasian direction chosen as a reference (Fig. 11), from one site to another. The interpretation of these directions is difficult since the palaeomagnetic and geochronological data available at the moment are insufficient for determining the component which in these apparent rotations is due to the evolution in time of the deformations at a given point, and that due to spatial variations of these deformations at a given period. In effect the comparison is made between directions obtained for formations of different ages situated in separate regions. We may, however, propose two provisional conclusions:

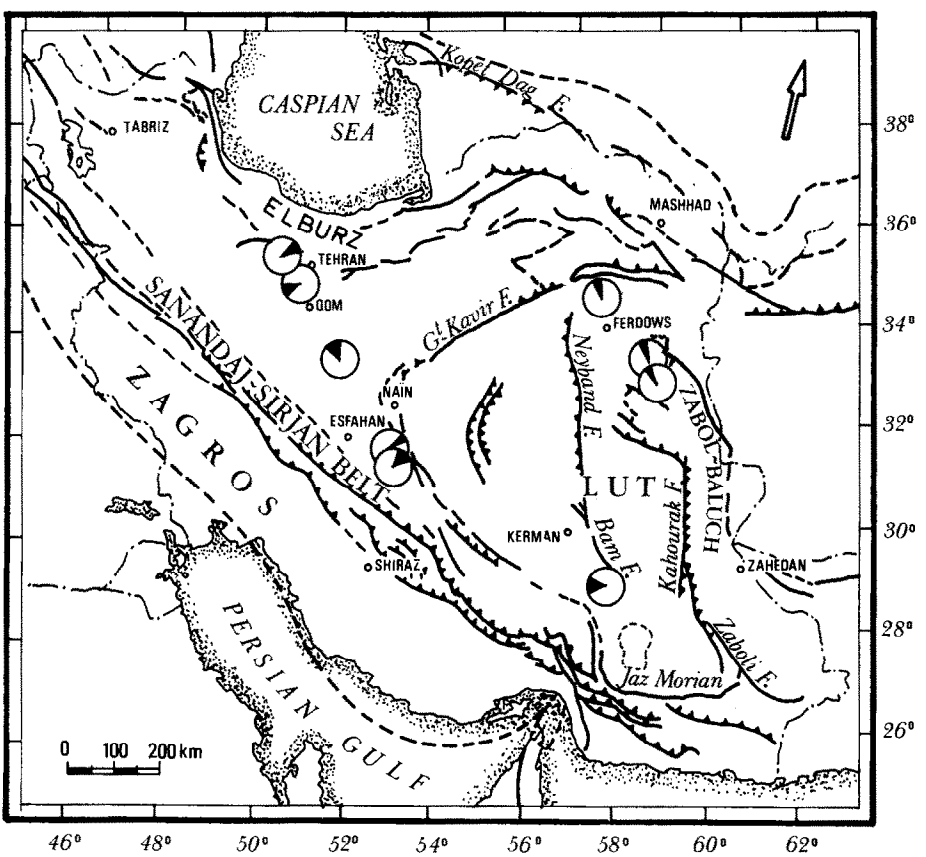


Fig. 11. Magnetic declination of Tertiary units from Iran (Table 3). The white arrow in the upper right corner indicates the mean Palaeogene direction for Eurasia.

(1) In eastern Iran, the north of the Zabol–Baluch zone and of the Lut block may have undergone, since the Palaeocene or the Eocene, a minor anticlockwise rotation with respect to the Eurasian reference direction.

(2) Moving towards the Zagros, that is to say closer to the contact zone of the Arabian plate, the rotations undergone by the Palaeogene formations become considerable, with the magnitude and direction varying from one nearby site to another (with, however, the exception of very close sites such as those of the Kuh-e-Shabadum and the Kuh-e-Kaleh-e-Kargushi). The data presently available are obviously insufficient for the definition of crustal blocks and their limits. The variety of the movements which have been evidenced do, however, seem to imply a structural model where numerous secant faults define relatively small-sized crustal compartments. A sliding effect along one of these faults may therefore result in rotations of different directions in the blocks situated on either side of the fault. The observed relative rotations could in our case result mainly from a large-scale movement between Iran and Arabia along the Zagros thrust zone (Brunn, 1976; Tapponnier, 1977; Berberian and King, 1981).

CONCLUSION

The palaeomagnetic, geochronological and petrological studies reported here, although fragmentary, nonetheless allow us to clarify some points regarding the geodynamic evolution of Iran during the Cainozoic:

(1) The K–Ar dates obtained indicate that the presumed Eocene magmatism of central Iran in fact covers a time span stretching from the Palaeocene (60 m.y. for the Bajestan–Gonabad andesites) to the Miocene (19 m.y. for the Hassan–Abad rhyodacites). Also, the Karaj dioritic intrusion (the central Elburz) is not of Eocene but Oligocene age.

The metamorphic retromorphosis evidenced by the petrological study in two Eocene volcanic series from Central Iran s.s. (Kuh-e-Shabadum andesites and Kuh-e-Kaleh-e-Kargushi shoshonites), which are transformations attributable to Amidi's (1975) static metamorphism, do not allow the determination of a K–Ar age for these lavas. The ages obtained correspond to the end of the metamorphism which is therefore dated to around 24 m.y., in agreement with earlier concepts (Amidi, 1975).

(2) As a general rule, the mean magnetic declination of the Palaeogene rocks studied is very different from one region to another, which proves the existence of large-scale relative movements. This prevents the calculation of a Palaeogene palaeomagnetic pole for Central Iran as a whole. Similarly, it does not appear possible to determine the apparent drift curve, before the Tertiary, of a palaeomagnetic pole which would be attributed to central Iran. The palaeomagnetic data obtained from the oldest rocks distributed in different zones of Iran must be corrected for relative movements of Alpine age, before being compared against each other. Unfortunately, the results obtained up to now do not yet allow the definition

of the limits and individual evolution of the different elements of the Iranian puzzle during the Cainozoic.

(3) The notion of a Persian plate, by definition rigid, is not applicable to the Cainozoic period. The reduction in width of the Iranian crust since the beginning of the collision with the Africo-Arabian plate, while perhaps resulting in a thickening of the crust (Berberian and King, 1981), is also accompanied by wide differential crustal rotations east of the Zagros. The data currently available suggest that these rotations are greater, and the crustal blocks smaller near the Zagros, than in the northeast of Iran. It would therefore seem reasonable to connect these important deformations with the sliding of continental Central Iran with respect to Arabia since the Eocene, a sliding whose direction and orientation would appear to have varied during the Cainozoic (Tapponnier, 1977). This interpretation is, however, hypothetical and new palaeomagnetic, geochronological and petrological studies will be necessary to a realistic mechanical model of this collision zone.

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